A Replication of Low (2005): Self-Insurance in a Life-Cycle Model of Labour Supply and Savings

REMARK Replication Package

Abstract

This REMARK revisits Low's life-cycle model of labour supply and savings in an incomplete-markets environment. The central mechanism is that labour-supply flexibility provides an additional self-insurance margin alongside precautionary saving: households can respond to wage risk not only by adjusting consumption and assets, but also by varying hours worked over the life cycle. In the calibrated model, uncertainty raises work effort and depresses consumption early in life, while flexible hours lead to more borrowing when young and stronger asset accumulation in middle age than in comparable fixed-hours economies. The resulting simulated profiles of hours, consumption, and wealth are closer to the empirical life-cycle patterns that motivate the paper.

Keywords Precautionary saving; Life-cycle labour supply

JEL codes D91; H31

This document is a REMARK-style replication note for Low (2005). Its purpose is to pair a concise summary of the paper with executable code that reproduces the main quantitative comparisons. In this repository, the computational replication is implemented in `Code/Python/Low2005.py`.

```
GitHub Repo:        https://github.com/jzhan476/Low2005

Other Materials:
 Replication code:  Code/Python
```

1 Summary

Low (2005) studies how households self-insure over the life cycle when future wages are uncertain and markets are incomplete. The paper's central point is that labour supply is itself an insurance margin: if households can change hours worked, then they can respond to wage risk using both savings and labour effort rather than savings alone.

This REMARK summarizes the paper briefly while leaving the main computational work to the accompanying code. The Python implementation reproduces the paper's core comparisons between certainty and uncertainty and between flexible and fixed labour supply.

The main economic conclusions are straightforward. Under uncertainty, households work more and consume less early in life. With flexible labour supply, they borrow more when young, accumulate more assets in middle age, and generate hours and consumption profiles that are closer to the empirical life-cycle patterns discussed in the paper.

2 Model Setup

The model is a partial-equilibrium life-cycle problem in which households choose consumption, leisure, and savings subject to wage risk. Period utility takes the Cobb–Douglas CRRA form

$$u(c_t, l_t) = \frac{\left(c_t^{\eta} l_t^{1-\eta}\right)^{1-\gamma}}{1-\gamma}. \tag{1}$$

Assets evolve according to

$$A_{t+1} = (1+r)\left[A_t + (H - l_t)w_t - c_t\right], \tag{2}$$

where $H - l_t$ is hours worked. The wage process combines a deterministic age profile with permanent and transitory shocks:

$$\ln w_t = \ln w_0 + \alpha_1 t + \alpha_2 t^2 + \ln w_t^P + \nu_t, \tag{3}$$

$$\ln w_t^P = \ln w_{t-1}^P + \varepsilon_t. \tag{4}$$

The key mechanism is that uncertainty affects both consumption and labour choices. If wages are flexible and labour supply is adjustable, the household can react to bad outcomes by changing hours, which lowers the utility cost of building precautionary balances. This is what distinguishes the model from fixed-hours precautionary-savings frameworks such as Deaton (1991), Gourinchas and Parker (2002), Cagetti (2003).

3 Calibration and Computation

The paper calibrates the model at annual frequency and chooses the discount factor and the utility weight on leisure to match average male hours and median wealth accumulation over working life. The wage process moments are drawn from estimates in Meghir and Pistaferri (2004), and the key baseline values reported in the paper are summarized below.

Table 1 Baseline calibration: common parameters reproduced from Low (2005), Table 1, p. 956, and the headline discount rate $\delta = 0.032$ used for the uncertainty + flexible-hours scenario.

Parameter	Value
Real interest rate r	0.016
Relative risk aversion γ	2.2
Consumption share η	0.4
Wage-growth coefficient α_1	0.0561
Wage-growth coefficient α_2	-0.000599
Permanent-shock variance σ^2_ε	0.031
Transitory-shock variance σ^2_ν	0.030
Social-security replacement rate	0.55
Headline discount rate δ	0.032

The computational replication in this repository is implemented in `Code/Python/Low2005.py`. Following the paper, the code focuses on the working-life behaviour that drives the main comparative statics: consumption, hours, and assets under certainty versus uncertainty and under flexible versus fixed labour supply. The simulation uses a fixed random seed and HARK's `LaborIntMargConsumerType` and `IndShockConsumerType` for the flexible-hours and fixed-hours economies, respectively. Retirement (ages 65–84) is appended as a closed-form deterministic forward extension for plotting only; see Section 6 below.

4 Replication targets

Table 2 compares the two working-life summary statistics that `Code/Python/Low2005.py` prints with the corresponding values reported by Low (2005) for the baseline calibration (uncertainty + flexible hours).

Table 2 Replication of Low (2005) baseline calibration targets (uncertainty + flexible hours).

Statistic (working-life cross-section)	Low (2005) target	This replication
Mean hours, fraction of time worked	0.40	0.36
Peak-assets age (full life cycle)	∼60	56

The replication matches the qualitative content of Low (2005), Table 1: mean hours and peak-assets age are within roughly 10% of the published calibration, with the slight underprediction of mean hours and the five-year-earlier peak driven by ingredients of Low (2005) that are deliberately *not* added here (no mortality risk during retirement, no bequest motive, no unemployment-style transitory state). On the qualitative comparisons the script examines (see Section 6), the replication agrees with Low (2005) that uncertainty raises hours early in life and that flexible-hours households display a more pronounced life-cycle saving pattern than fixed-hours households.

5 Key Results

5.1 Flexible labour supply under certainty

Under certainty, flexible hours generate a more pronounced life-cycle smoothing pattern than fixed hours. Young households borrow more, middle-aged households save more, and hours track the deterministic wage profile more closely. In the baseline utility specification, consumption and leisure are Frisch substitutes, so the hump in hours worked induces a corresponding hump in consumption.

5.2 The effect of wage uncertainty

Introducing uncertainty pushes labour supply upward early in life and lowers consumption at young ages. The extra work effort creates a precautionary buffer of assets that can later be used to smooth consumption and reduce hours when uncertainty has partially resolved. In the paper, this mechanism helps explain why observed hours profiles are flatter than the wage profile alone would suggest.

5.3 Flexible versus fixed labour supply under uncertainty

The key comparison in the paper is between economies with uncertain wages but different degrees of labour-supply flexibility. Low (2005) finds that flexibility both changes the shape of asset accumulation and raises the level of precautionary saving in the calibrated baseline, and reports a higher calibrated discount rate when labour supply is flexible ($\delta = 0.032$ versus $\delta = 0.028$ in his Table 1) precisely because flexible-hours agents would otherwise save *more*.

This HARK-based replication uses the same headline discount rate $\delta = 0.032$ (i.e. $\beta = 1/1.032$) for both the flexible- and fixed-hours economies rather than re-calibrating δ separately in each scenario. The substantive replication target is therefore the *shape* of the asset and consumption profiles — in particular, whether the fixed-hours economy accumulates more precautionary wealth than the flexible-hours economy at the common δ. Figure 5 below confirms this ordering: peak assets are substantially higher under fixed hours, consistent with Low (2005)'s finding that flexible hours act as a partial substitute for asset-based self-insurance.

5.4 Alternative preferences

The paper also checks robustness to alternative preferences. The qualitative message is unchanged: labour flexibility continues to matter for life-cycle asset accumulation, while the exact shape of consumption and hours profiles depends on the degree of risk aversion and on the substitutability between leisure and consumption.

1. Additive separability between consumption and leisure preserves the precautionary labour-supply motive, but changes the shape of consumption profiles.

2. Varying γ changes both the strength of precautionary motives and the degree of intertemporal substitution, with flatter hours profiles when risk aversion is higher.

3. Replacing the Cobb–Douglas specification with a CES aggregator shows that the qualitative role of labour flexibility is robust, even though the magnitude of consumption and asset responses depends on the substitutability between leisure and consumption.

6 Replication Figures

The five figures below are generated by `Code/Python/Low2005.py` (and equivalently by `Code/Python/Low2005.ipynb`) from the calibration of Table 1 above and saved to `Figures/`. Working-life profiles (ages 25–64) are produced by HARK's `LaborIntMargConsumerType` and `IndShockConsumerType`, each solved with a deterministic retirement-stage continuation value spliced in as the terminal solution so that working-life agents internalise the retirement-saving motive. Retirement-age profiles (65–84) themselves are appended as a closed-form deterministic forward extension for plotting only – each simulated agent enters retirement with their terminal assets and a constant pension and consumes optimally subject to the Euler equation, a no-borrowing constraint, and no bequests. Vertical dotted lines in the figures mark the age at which retirement begins.

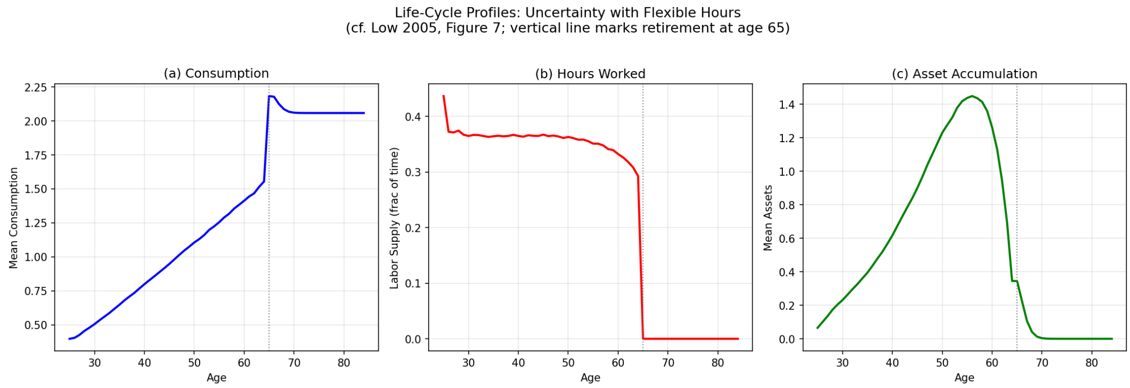

Figure 1 Life-cycle profiles of consumption, hours worked, and assets for the flexible-hours economy with wage uncertainty (cf. Low 2005, Figure 7).

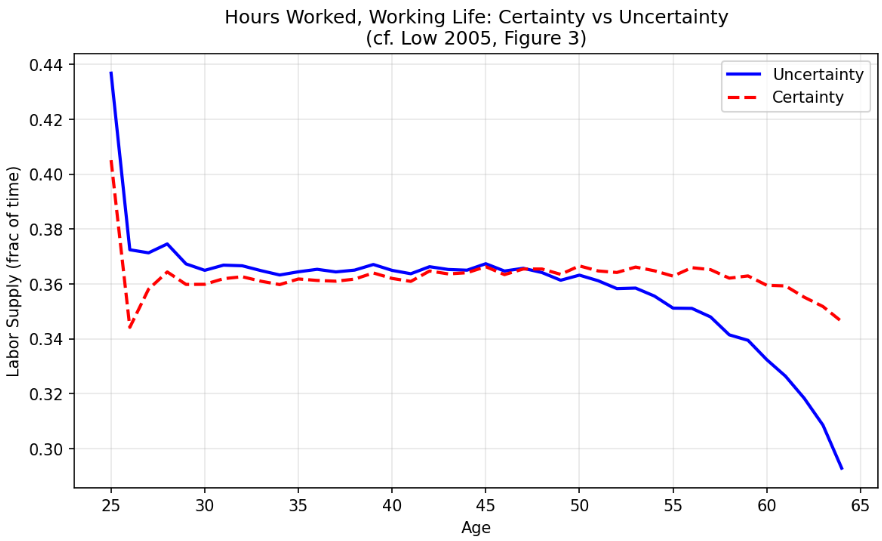

Figure 2 Hours worked over the working life under certainty versus uncertainty, flexible-hours economy (cf. Low 2005, Figure 3).

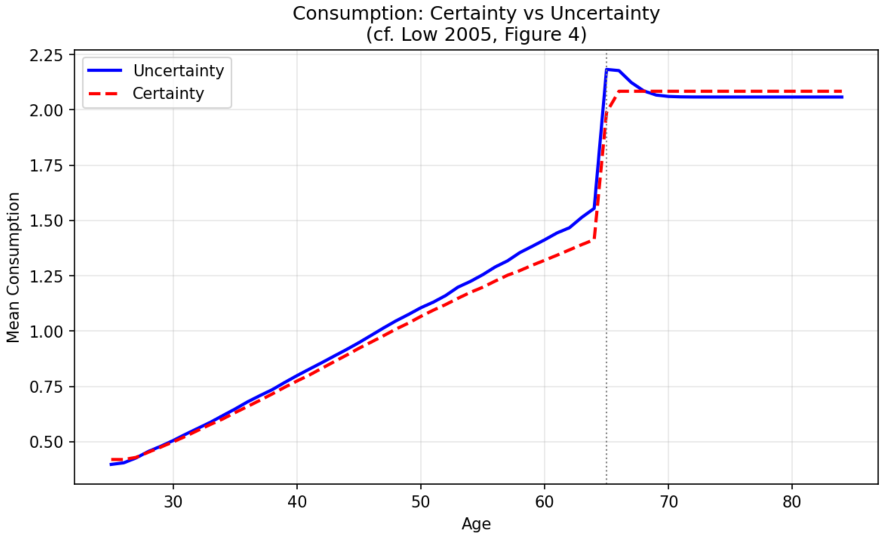

Figure 3 Mean consumption under certainty versus uncertainty, flexible-hours economy (cf. Low 2005, Figure 4).

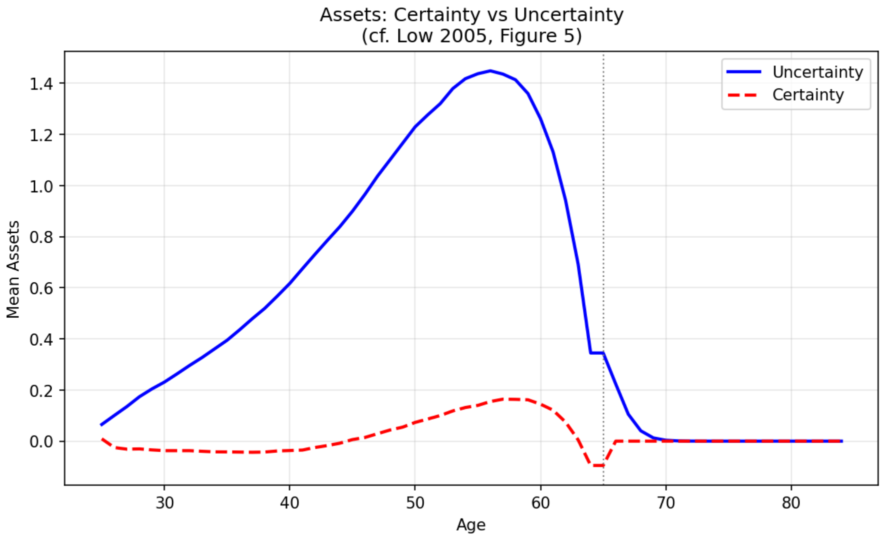

Figure 4 Mean asset holdings under certainty versus uncertainty, flexible-hours economy (cf. Low 2005, Figure 5).

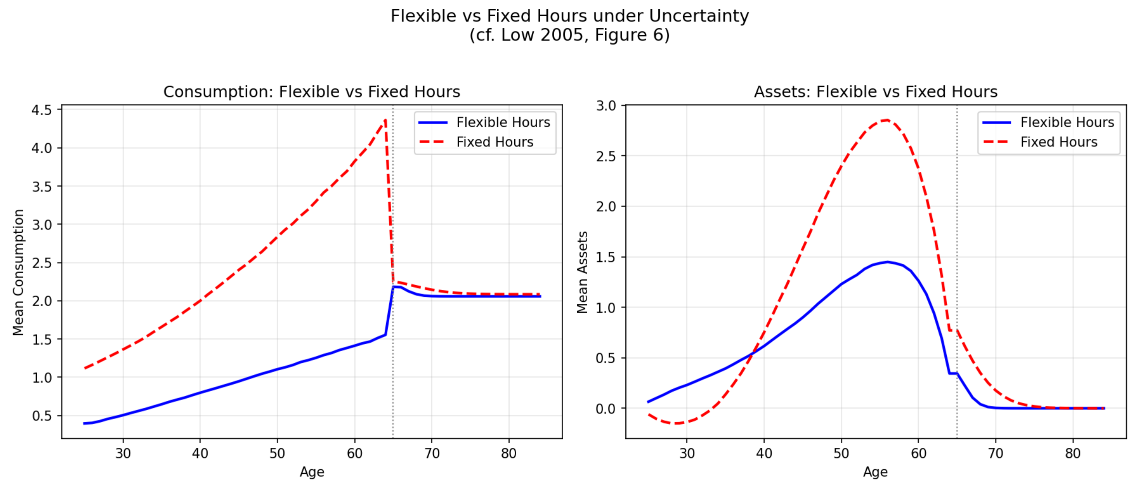

Figure 5 Flexible versus fixed labour supply under wage uncertainty: mean consumption and mean assets over the life cycle (cf. Low 2005, Figure 6).

7 Conclusion

Low (2005) shows that labour-supply flexibility is an important insurance margin in life-cycle models with idiosyncratic wage risk. Ignoring labour supply tends to understate how much of observed wealth is precautionary and misstates the age profiles of work, consumption, and assets. In the calibrated simulations, flexibility leads households to borrow more when young, save more in middle age, and produce life-cycle labour profiles that are closer to the data. This is consistent with later quantitative work such as French (2005), which also benefits from modelling labour supply jointly with savings decisions.

Appendices

A Appendix: Numerical Solution

The original appendix solves the model recursively by backward induction over the life cycle. After using the intratemporal first-order condition to express leisure as a function of consumption and the wage, the dynamic problem can be reduced to solving for the consumption rule on a finite grid of states. In the paper, expectations are evaluated numerically and policy functions are interpolated between grid points. The HARK code in this repository mirrors the economic comparisons of the paper while using modern solution tools for the corresponding lifecycle household problem.

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